Single-cell deciphering of epithelial-mesenchymal transition and endothelial-mesenchymal transition reveals cancer-associated fibroblast plasticity and diversity in HPV+ head and neck cancer

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Introduction

- Tumor microenvironment crucial for tumor progression in head and neck squamous cell carcinoma (HNSCC)
- HPV-positive HNSCCs have better prognosis due to enhanced immune surveillance
- Key processes: **epithelial-mesenchymal transition** (EMT) and **endothelial-mesenchymal transition** (EndMT), where cells lose their epithelial characteristics (cell-cell adhesion, polarity) and gaining mesenchymal traits (increased motility, invasiveness)
- EMT/EndMT causes formation of **cancer-associated fibroblasts** (CAFs), alters extracellular matrix (ECM) and promotes a pro-tumorigenic environment
- CAFs are heterogeneous, impacting tumor behavior based on phenotype and ECM composition

Objectives

- Investigate and identify subpopulations within the tumor microenvironment, including malignant epithelial cells, CAFs, endothelial cells, and transitioning EMT/EndMT cells
- Examine the roles of EMT and EndMT in tumor progression and their impact on cancer-associated fibroblast (CAF) formation and function
- Elucidate the influence of TGF-β signaling and associated pathways in promoting angiogenesis and regulating CAF activation and EndMT.
- Evaluate the presence and implications of partial EMT and epithelial-mesenchymal plasticity in tumor migration, metastasis, and treatment resistance

Methods

- Dataset provided by collaborators who conducted single-cell sequencing on tumor biopsy samples from HPV-positive oropharyngeal carcinoma patients in a phase 2 trial receiving ipilimumab and nivolumab with concurrent radiotherapy at baseline and after induction
- Analyzed subset of EMT-related cells at baseline

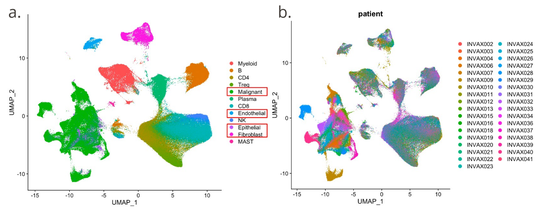


Fig. 1: **a.** UMAP plot of original dataset with subset of EMT-related cells selected for analysis **b.** UMAP plot of samples for each patient in trial

- Aligned sequencing reads, quantified gene expression, filtered cells based on quality. Applied normalization, identified variable genes, and performed dimensionality reduction using PCA and UMAP for clustering.
- Identified malignant cells through epithelial scoring, HPV gene expression, and copy number variation analysis.
- Conducted clustering and differential expression analysis
- Labeled major cell types with canonical markers. Examined cells with high EMT scores and CAF markers to identify key markers and distinct fibroblast subpopulations.
- Mapped cell trajectories to explore potential transformations. Analyzed RNA velocity for insights into transition pathways and identified significant cell-cell interactions and transition-driver genes. Ran copy-number analysis to identify changes in malignant cells

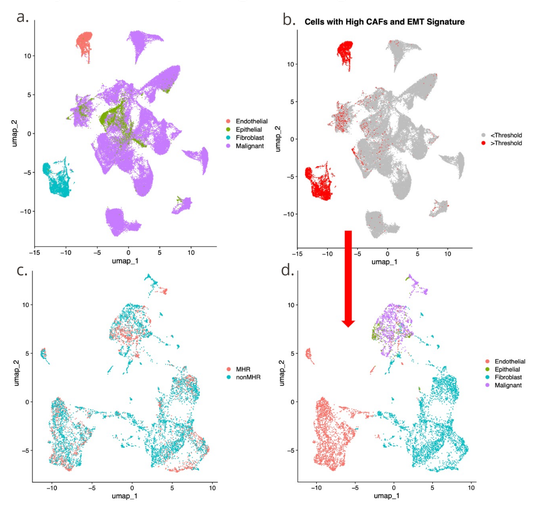


Fig. 2: **a.** UMAP plot of EMT-related subset of cells **b.** UMAP plot of cells that express a high CAFs and EMT gene signature, scored by AUCell global k1 threshold **c.** UMAP plot of selected cells colored by cell label **d.** UMAP plot of selected cells colored by response

Results

- Uncovered cancer-associated fibroblast diversity with specific markers defining subtypes: TAGLN+ and COL11A1+ fibroblasts activate CAFs through TGF-β signaling (NF-κB/IGFBP2 axis), POSTN+/GREM1+ CAFs are involved in immune response barriers and regulate EndMT, and for CXCL13+ fibroblasts, CAFs have been shown to drive CXCL13 production in activated T cells via TGF-β
- Endothelial-fibroblast transition cells express immune markers (PTPRC, CD52), potential immune cell crosstalk impacting immunotherapy response
- Malignant transitioning cells express LAMC2, LAMB3, previously identified markers of meta-stable partial EMT cells, while malignant epithelial cells express CEACAM5 and CEACAM6, which play role in cell adhesion, cell invasion and tumor progression

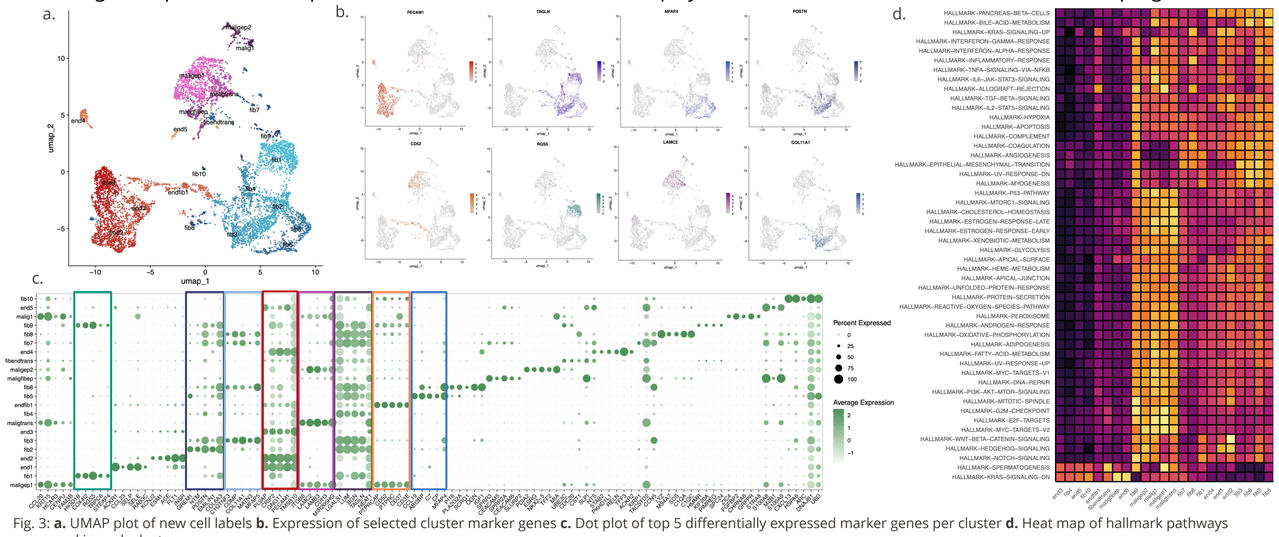


Fig. 3: **a.** UMAP plot of new cell labels **b.** Expression of selected cluster marker genes **c.** Dot plot of top 5 differentially expressed marker genes per cluster **d.** Heat map of hallmark pathways expressed in each cluster

RNA Velocity

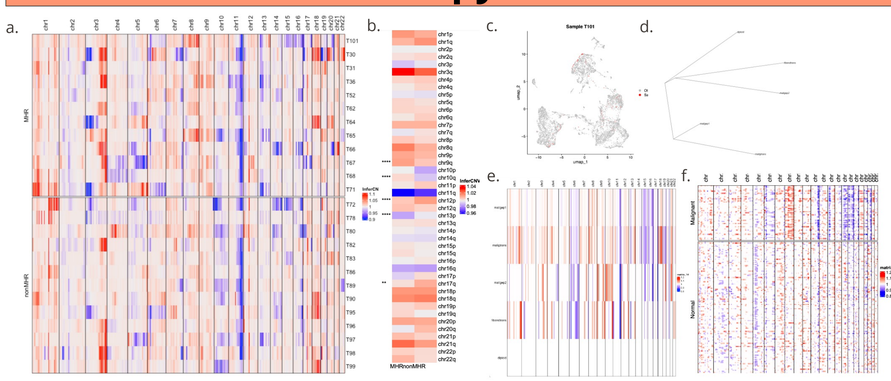
Fig. 4: **a.** RNA velocity plot generated with scVelo dynamical model **b.** Cells colored according to inferred latent time, which estimates the cell's position along the developmental trajectory **c.** Top likelihood driver genes for selected clusters **d.** Top likelihood driver genes across latent time

- Velocity plot shows stem-like endothelial cell differentiation to CAFs
- Undifferentiated malignant cells in low latent time transition to high latent time CAFs
- Driver genes include EMT-related genes, transcription factors TP63, BASP1

Cell Trajectories

- State 2 branch contains mix of multiple cell types in partial-EMT state, found in EMT/EndMT transition
- LAMB3 and LAMC2 are previously identified markers of partial-EMT, increase across pseudotime in state 2
- Enrichment of gene modules across branch-dependent pseudotime shows distinct functions (angiogenesis, ECM, etc.)

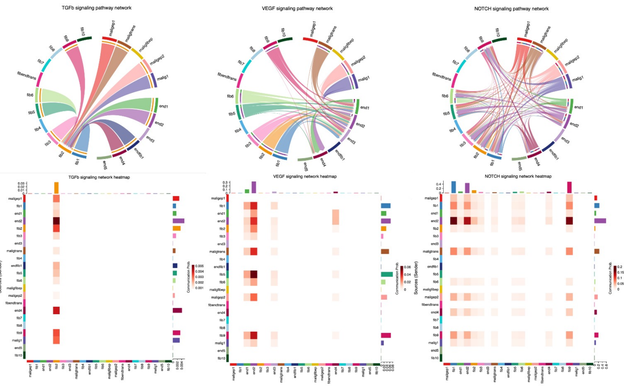
Copy Number Variation



- Significant DNA differences between normal and malignant cells, particularly on chromosomes 11q and 3q for MHR and nonMHR
- Increased genome variation in chr 3q, functions include epithelial structure maintenance, Notch pathway regulation
- CNV profiles and tree of normal vs malignant cell subtypes show possible lineages

Fig. 5: **a.** Heat map of CNV per sample **b.** Combined CNV across all samples **c.** Location of cells in sample T101 **d.** CNV tree of malignant cell types **e.** Heatmap of CNV per malignant cell type **f.** Heatmap of CNV between normal and malignant samples

Cell-cell Interactions



- Endothelial TGF-β1 signaling to fibroblasts induces EndMT and promote VEGF transcription and secretion in via Smad pathway, enhancing angiogenesis
- End2 subset also involved in VEGF signaling (receives signals from fibroblasts) and NOTCH signaling (sends signals to fibroblasts), implicated in malignant transformation
- Key markers in End2 include ESM1 (upregulated in cancer EndMT, INSR (promotes angiogenesis), needs further analysis

Conclusions

- Complex interactions and transitions among endothelial, epithelial, and fibroblast populations in HPV-positive head and neck cancer drive tumor progression
- Notable cell plasticity was identified, including fibroblast-like cells in a meta-stable partial EMT state with cell-specific markers such as LAMC2 and LAMB3, associated with increased metastasis and poor prognosis
- TGF-β signaling is critical, influencing VEGFA signaling and angiogenesis, essential for CAF activation and EndMT regulation
- Notch signaling from ESM1+/INSR+ endothelial cell population implicated in malignant cell transformation
- Transcription factors TP63, BASP1 may drive malignant EMT
- Targeting these pathways and markers is vital for overcoming resistance and improving patient outcomes

Future Work

- Investigate specific expression patterns of driver genes to understand their roles in tumor progression
- Further explore copy number variation to identify additional chromosomal alterations and their implications
- Conduct a comparative analysis of non-major histologic responders (nonMHR) versus major histologic responders (MHR) to identify biomarkers and mechanisms associated with treatment efficacy
- Analyze transcription factors to delineate their regulatory roles in epithelial-mesenchymal transition and cancer-associated fibroblast function

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